
Brain MRI Tumor Classification with ROI Segmentation

Project Proposal (Implementation Focused)

Course: ECEN 766: Algorithms in Structural Bioinformatics

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Primary Tooling: MATLAB (Image Processing Toolbox, Statistics & Machine Learning Toolbox)

Abstract

Brain tumors are life-threatening and require early detection and careful follow-up. Magnetic Resonance Imaging (MRI) provides rich anatomical detail, but manual interpretation is time-consuming and may vary between radiologists. This project proposes a practical MATLAB-based pipeline for brain MRI tumor classification using classical image processing and machine learning techniques.

An existing wavelet-feature classification approach is adopted as the baseline method. The system is extended by introducing a tumor Region-of-Interest segmentation stage before feature extraction. Two pipelines are implemented and compared. The baseline pipeline performs preprocessing followed by feature extraction on the full MRI image and classification. The proposed pipeline performs preprocessing, ROI segmentation, feature extraction from the ROI, and classification. The impact of segmentation is evaluated using confusion-matrix metrics including accuracy, precision, recall, and F1-score under cross-validation or repeated hold-out evaluation. The focus of this work is on an implementable, interpretable, and reproducible system rather than highly complex models.

1 Introduction

The baseline idea of this project is to extract wavelet-based features from the full MRI image and perform classification. However, full-image features can be influenced by non-tumor regions such as healthy tissue, skull structures, and background intensity variations. These regions may dilute tumor-specific information and reduce classification reliability.

To address this issue, the proposed improvement is to introduce a lightweight tumor Region-of-Interest (ROI) segmentation step before feature extraction. The goal is to isolate the tumor region and compute features only from the relevant area. The final deliverable is the implementation of both the baseline and proposed pipelines in MATLAB and a comparison of their performance using the same evaluation protocol.

2 Problem Definition and Significance

The goal of this project is to build an automated pipeline capable of classifying brain MRI images based on tumor presence or absence. Given a labeled dataset of MRI images, the system must preprocess the images, extract meaningful features, and predict the correct class label. The central research question explored in this project is whether adding an explicit tumor ROI segmentation step improves classification performance compared to extracting features from the entire MRI image.

This problem is important because MRI images contain large non-tumor regions that may interfere with feature extraction. Global features computed across the entire image may be dominated by

healthy tissue or background patterns rather than tumor characteristics. Introducing a reliable and lightweight ROI segmentation step may improve robustness while remaining feasible for student-level implementation.

The project also provides practical exposure to a complete applied machine learning workflow including preprocessing, segmentation, feature extraction, classification, and performance evaluation.

3 Illustrative Clinical Scenario

Consider two axial MRI slices obtained from different patients. Both slices contain bright regions in the brain image. One region corresponds to normal cerebrospinal fluid near the ventricles, while the other represents a small tumor located close to the skull. A classifier that extracts wavelet features from the entire MRI slice may interpret both bright regions as similar textures and may incorrectly classify the tumor case as a non-tumor case. By introducing a segmentation step, the system explicitly

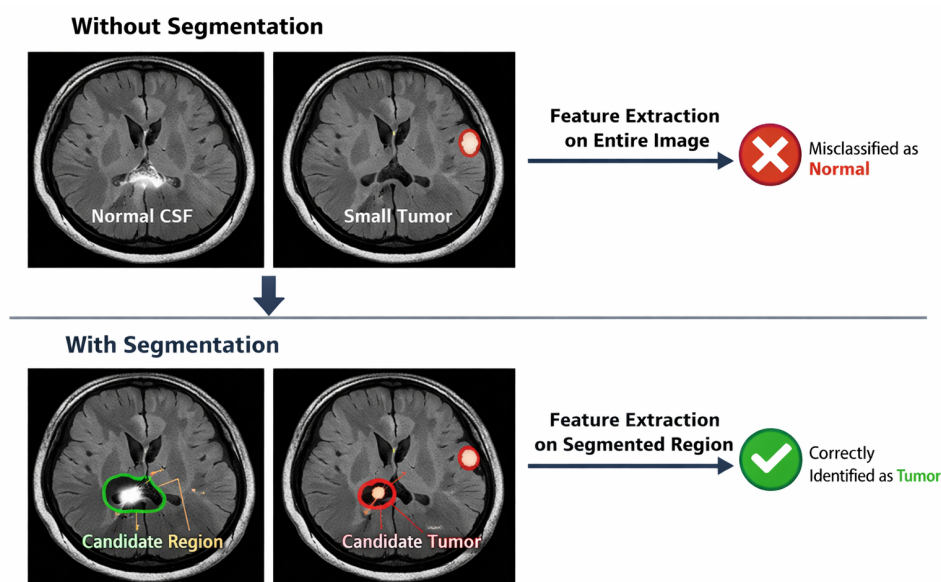


Figure 1: Effect of segmentation in MRI tumor detection. Segmentation isolates abnormal regions before feature extraction, improving classification reliability.

searches for compact abnormal regions and isolates them as candidate tumor areas. Feature extraction is then performed only on this segmented region. In a clinical workflow such a system would not replace a radiologist but could highlight suspicious areas and provide an automated second opinion indicating whether the scan resembles tumor or non-tumor patterns learned from the dataset.

4 Current Methods and Baseline

The baseline approach used in this project follows the method proposed by Mishra (2010), which extracts wavelet packet features from MRI images and uses an artificial neural network classifier for abnormality detection. Wavelet transforms decompose the image into multiple frequency subbands, allowing the representation of texture and edge patterns at different spatial scales. Statistical properties derived from these subbands form a compact feature vector that can be used for classification.

Although this method provides a strong baseline for medical image classification, it does not explicitly isolate the tumor region. All pixels in the image, including skull regions, normal brain tissue, and background areas, contribute to the feature vector. As a result, the classifier may learn patterns that are unrelated to tumor characteristics.

The main gap addressed in this project is therefore the absence of explicit tumor region isolation prior to feature extraction.

5 Proposed Method

The proposed method extends the baseline pipeline by introducing a tumor ROI segmentation stage before feature extraction. The high-level pipeline consists of MRI input, preprocessing, ROI segmentation, feature extraction, classification, and final prediction.

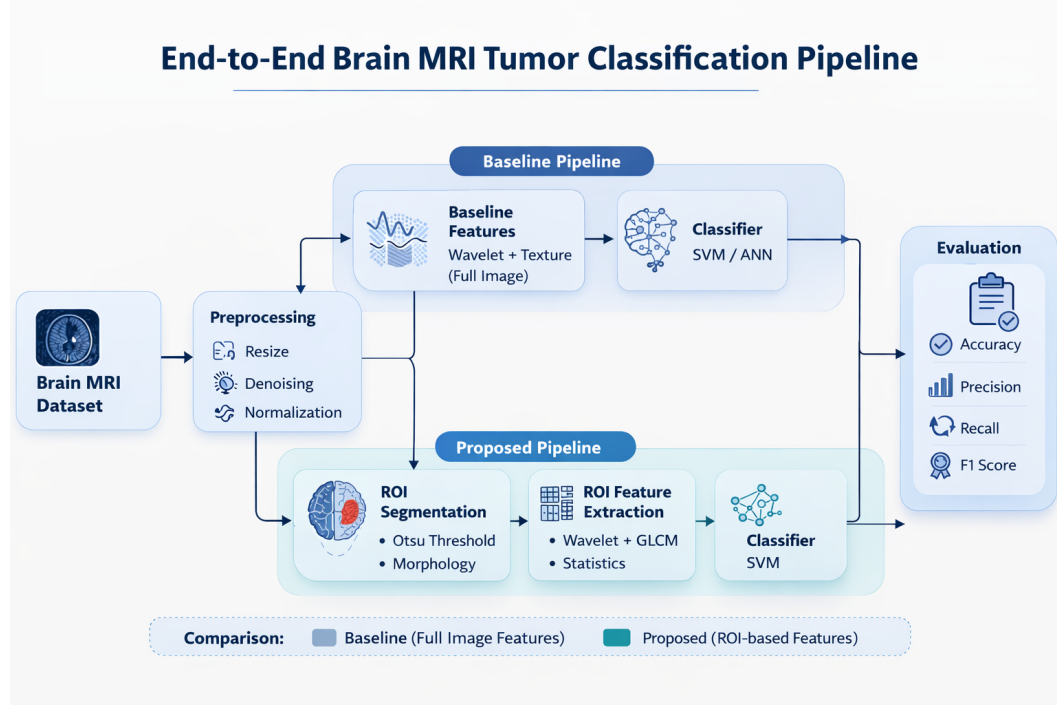


Figure 2: End-to-end brain MRI tumor classification pipeline comparing the baseline full-image feature approach with the proposed ROI-based segmentation framework.

5.1 Data

The system will use a publicly available labeled brain MRI dataset containing at least two classes. Each two-dimensional MRI slice will be treated as a separate sample with its corresponding class label. The initial implementation focuses on binary classification such as tumor versus non-tumor or benign versus malignant depending on dataset availability and labeling quality.

5.2 Preprocessing

All images will be resized to a fixed spatial resolution to ensure consistent feature extraction. If necessary, images will be converted to grayscale. Noise reduction will be performed using median filtering or Wiener filtering. Intensity normalization or contrast adjustment may also be applied to improve image quality. A small sample of images will be visualized before and after preprocessing to verify correctness.

5.3 ROI Segmentation

The ROI segmentation module is designed to be lightweight and feasible to implement in MATLAB without requiring large-scale training data. The goal is not perfect pixel-level segmentation but rather the identification of a reasonable candidate tumor region that improves classification.

First, Otsu thresholding is applied to the preprocessed grayscale image to obtain a binary foreground mask. Second, morphological operations are used to remove small isolated regions and fill holes within the mask. Third, connected component analysis is used to select the most likely tumor candidate region based on simple heuristics such as component size or location. If necessary, watershed segmentation on gradient images may be explored as an optional refinement.

5.4 Feature Extraction

Feature extraction will be performed differently for the baseline and proposed pipelines. In the baseline system, features are extracted from the entire preprocessed MRI image. In the proposed system, features are extracted only from the segmented tumor region.

The features include wavelet-based statistics obtained from discrete wavelet transform subbands, texture descriptors derived from gray-level co-occurrence matrices, and basic statistical features such as mean, variance, and entropy. If the feature dimensionality becomes large, principal component analysis may be used to reduce dimensionality.

5.5 Classification

Support Vector Machines will be used as the primary classifier due to their strong performance on moderate-sized feature vectors and relatively simple training procedure. Both linear and radial basis function kernels will be evaluated. Additional models such as shallow neural networks may be explored if time permits.

6 Experimental Evaluation

The experimental design focuses on isolating the effect of ROI segmentation. Two pipelines will be compared using the same dataset, training protocol, and classifier configuration. The baseline pipeline performs preprocessing followed by full-image feature extraction and SVM classification. The proposed pipeline performs preprocessing, ROI segmentation, feature extraction on the segmented region, and SVM classification.

Five-fold cross-validation will be used when the dataset size permits. Otherwise repeated hold-out evaluation will be used. Stratified splitting will ensure balanced class distributions across training and testing sets. If patient identifiers are available, images from the same patient will be kept within the same fold to prevent data leakage.

Performance will be evaluated using confusion matrices, accuracy, precision, recall, and F1-score. When probability outputs are available, receiver operating characteristic curves and area under the curve may also be reported.

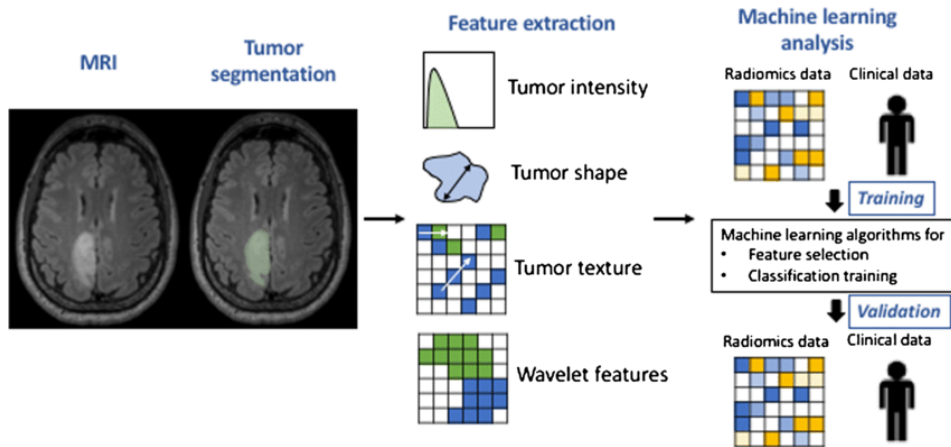


Figure 3: Experimental evaluation setup comparing the baseline full-image feature pipeline with the proposed ROI-based segmentation pipeline. Both pipelines use identical datasets, training protocols, and SVM classifiers to isolate the impact of ROI segmentation on classification performance.

7 Project Timeline

The project will be implemented in stages to ensure steady progress. The first stage focuses on environment setup, dataset acquisition, and preprocessing implementation. The second stage implements the baseline classification pipeline and produces initial evaluation results. The third stage develops the ROI segmentation module and validates it visually. The fourth stage integrates segmentation into the classification pipeline and performs comparative evaluation. The final stage focuses on polishing the system, generating figures and tables, and preparing the final report.

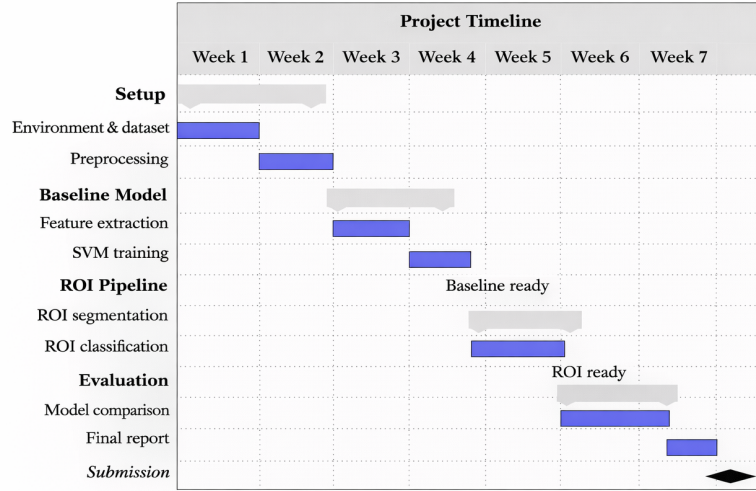


Figure 4: Seven-week project timeline for the Brain MRI tumor classification system.

8 Computing Resource Estimation

The proposed pipeline is designed to run on a standard laptop. The dataset storage requirement is expected to be between approximately 0.5 GB and 5 GB depending on the selected dataset. Memory requirements are estimated at 8 to 16 GB RAM for processing feature matrices and running experiments. Preprocessing, segmentation, and feature extraction are expected to take only a few seconds per image, while SVM training will require minutes depending on dataset size. A GPU is not required unless deep learning models are introduced as optional extensions.

9 Contingency Plan

If segmentation performance proves unstable, the simplest robust segmentation method consisting of Otsu thresholding followed by morphological cleanup will be used. Any limitations will be documented using visual examples in the final report. If the dataset is small or imbalanced, stratified cross-validation will be used and class-specific performance metrics will be reported. If time constraints arise, the project will prioritize delivering a clear comparison between the baseline and proposed pipelines using a single classifier and segmentation method.

References

R. Mishra. MRI based brain tumor detection using wavelet packet feature and artificial neural networks. Proceedings of the International Conference and Workshop on Emerging Trends in Technology (ICWET), 2010.